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CONTROL SYSTEM

Abstract

A control system for growing plant matter such as algae, the control system having a controller for controlling a plurality of LED lights, wherein the controller is adapted to control the spectrum and intensity of the LED lights. Preferably the controller is also adapted to control a flickering frequency of the LED lights.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. application Ser. No. 17/287,570, filed Apr. 22, 2021, which is a National Stage filing of PCT/AU2019/051152, filed Oct. 22, 2019, each of which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present invention relates to a control system for growing algae. Reference will be made in the specification to the use of the invention with respect to growing algae. The patent specification describes this use, but it is by way of example only and the invention is not limited to this use.

BACKGROUND

[0003] Commercial cultivation of Algae is used to produce food ingredients, food colorants and dyes, bioplastics, pharmaceuticals, and algal fuel amongst others. Most Algae requires water, nutrients, a carbon source and light to grow.

[0004] How all these elements are administered determines the quality and quantity of the algae. There are a large number of combinations of how these elements can be administered to grow different types of algae.

[0005] Light is an important element in growing algae. Direct sunlight is too strong for most algae. However, direct sunlight is often best for strong growth as the algae underneath the surface is able to utilize the less intense light created from the shade above.

[0006] One prior art solution to control the exposure of algae to light is to agitate the algae.

SUMMARY

[0007] It is an object of the present invention to overcome or at least alleviate one or more of the above mentioned problems with growing algae and/or provide the consumer with a useful or commercial choice.

[0008] In one aspect, the present invention broadly resides in a control system for growing plant matter having

[0009] a controller for controlling a plurality of LED lights, wherein the controller is adapted to control the spectrum and intensity of the LED lights.

[0010] Preferably, the controller is adapted to control a flickering frequency of the LED lights. Preferably the flickering frequency is a frequency of on/off cycles of the LED light. Preferably the controller is adapted to control the flickering frequency in a range of 2 ms to 5,000 ms. More preferably the controller is adapted to control the flickering frequency in a range of 25 ms to 1,000 ms. In another embodiment, the controller is adapted to control the flickering frequency in a range of 2 ms to 500 ms. In a further embodiment, the controller is adapted to control the flickering frequency in a range of 10 ms to 100 ms.

[0011] Preferably the controller includes a communication module. The communication module is preferably adapted to communicate with a remote device. In one embodiment, the remote device is a mobile device such as a mobile telephone. In another embodiment, the remote device is a computer. In a further embodiment, the remote device is a server. Preferably the controller communicates with the server via the internet.

[0012] In one embodiment, the controller is adapted to communicate with more than one remote device.

[0013] The remote device is preferably adapted to set the spectrum and intensity of the LED lights.

The remote device is preferably adapted to set the flickering frequency of the LED lights.

[0014] Preferably the controller is adapted to communicate with one or more sensors. Preferably the one or more sensors provide data on the growth of the plant matter.

[0015] Preferably the one or more sensors include one or more gas sensors. Preferably the one or more gas sensors include a carbon dioxide, hydrogen and/or oxygen sensor. In one embodiment, the one or more gas sensors are adapted to sense dissolved gas in growing medium used to grow plant matter such as algae. The growing medium is preferably a liquid such as water.

[0016] Preferably the one or more sensors include a sensor to determine plant growth. Preferably the one or more sensors includes a spectrophotometer to determine plant growth. In one embodiment, the one or more sensors includes a spectrophotometer to determine algae cell density in liquid culture.

[0017] Preferably the one or more sensors include a temperature sensor. Preferably the temperature sensor senses a temperature of the growing medium.

[0018] Preferably the one or more sensors includes a pH sensor to measure the pH of the growing medium.

[0019] Preferably the one or more sensors include a salinity sensor. Preferably the salinity sensor measures the salinity of the growing medium.

[0020] Preferably the one or more sensors include a nitrate sensor. Preferably the nitrate sensor measures the concentration of nitrates in the growing medium.

[0021] Preferably the one or more sensors include a cell count sensor. Preferably the cell count sensor measures the quantity of plant matter cells in a defined volume.

[0022] Preferably the one or more sensors include a cell health sensor. Preferably the cell health sensor provides data on the health of plant matter cells.

[0023] The controller is preferably adapted to receive data from the one or more sensors. The controller is preferably adapted to send the data from the one or more sensors to the remote device. The controller is preferably adapted to send data on the spectrum and intensity of the LED lights to the remote device. The controller is preferably adapted to send data on the flickering frequency of the LED lights to the remote device. The remote device is preferably adapted to optimise the spectrum and intensity of the LED lights based on the data from the one or more sensors. Preferably the remote device correlates data from the one or more sensors with data on the spectrum and intensity of the LED lights. Preferably the remote device correlates data from the one or more sensors with data on the flickering frequency of the LED lights. Preferably the remote device will determine the spectrum and intensity of the LED lights that correspond to desirable data from the one or more sensors. Preferably the remote device will set the spectrum and intensity values in the controller to the determined spectrum and intensity of the LED lights. Preferably the remote device will determine the flickering frequency of the LED lights that corresponds to desirable data from the one or more sensors. Preferably the remote device will set the flickering frequency in the controller to the determined flickering frequency of the LED lights.

[0024] Preferably the remote device will correlate the type of plant matter being grown with the data on the spectrum and intensity of the LED lights. Preferably the remote device will correlate the type of plant matter being grown with the data from the one or more sensors. Preferably the remote device will correlate the type of plant matter being grown with the data on the flickering frequency of the LED lights.

[0025] In one embodiment, a remote device defines the type of plant matter being cultivated and communicates this data with the controller, the controller communicates this data with a further remote device and downloads from the further remote device spectrum and intensity settings of the LED lights which correlate with the type of plant matter being cultivated.

[0026] Preferably the remote device will correlate desired traits of plant matter being grown with the data on the spectrum and intensity of the LED lights. Preferably the remote device will correlate desired traits of plant matter being grown with the data from the one or more sensors.

Preferably the remote device will correlate desired traits of plant matter being grown with the data on the flickering frequency of the LED lights.

[0027] In one embodiment, a remote device defines the desired traits of plant matter being cultivated and communicates this data with the controller, the controller communicates this data with a further remote device and downloads from the further remote device spectrum and intensity settings of the LED lights which correlate with the desired traits of plant matter being cultivated.

[0028] In another embodiment, the remote device is used to manually set the spectrum and intensity settings of the LED lights in the controller.

[0029] Preferably the controller is adapted to control the spectrum of the LED lights in the range of 200 nm to 800 nm. More preferably the controller is adapted to control the spectrum of the LED lights in the range of 380 nm to 750 nm.

[0030] Preferably the LED lights to provide the plant matter with light.

[0031] Preferably the controller is adapted to control each of the plurality of LED lights individually. In one embodiment the plurality of LED lights are divided into zones, with the controller adapted to control each zone individually. In one embodiment, each zone is used to cultivate different plant matter. In another embodiment, each zone is used to cultivate plant matter at a different growth stage. In a further embodiment, each zone is used to develop different traits in the plant matter.

[0032] Preferably the plant matter is algae.

[0033] Preferably the control system includes the LED lights.

[0034] Preferably the control system includes the one or more sensors.

[0035] In one embodiment, the control system includes the remote device.

[0036] In another aspect, the present invention broadly resides in an algae cultivation control system, the control system having a controller for controlling a plurality of LED lights to provide the algae with light, wherein the controller is adapted to control the spectrum and intensity of the LED lights.

[0037] In a further aspect, the present invention broadly resides in a control system for growing algae having a controller for controlling a plurality of LED lights, wherein the controller is adapted to control the spectrum and intensity of the LED lights.

[0038] In another aspect, the present invention broadly resides in in a control system for growing algae having a plurality of LED lights to provide the algae with light; and a controller for controlling the plurality of LED lights, wherein the controller is adapted to control the spectrum and intensity of the LED lights.

[0039] Preferably the control system includes one or more sensors. Preferably the one or more sensors provide data on the growth of the algae.

[0040] Preferably the control system includes a communication module. More preferably, the controller includes the communication module. The communication module is preferably adapted to communicate with a remote device. The communication module is preferably adapted to enable communication between the remote device and the controller.

[0041] In one embodiment, the remote device is adapted to communicate with one or more further control systems. Preferably the remote device is adapted to communicate the settings for the spectrum and intensity of the LED lights which have resulted in desirable traits in the plant matter to the one or more further control systems.

[0042] Preferably the remote device is adapted to communicate additional settings to the control system and or the one or more further control systems. Preferably the additional settings include one or more of flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added. Preferably the corresponding controller is adapted to control one or more pumps, valves, solenoids, heaters, or the like, to control the flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients

added. Preferably the corresponding controller controls one or more pumps, valves, solenoids, heaters, or the like, to control the flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added, according to the settings received from the remote device.

[0043] In a further aspect, the present invention broadly resides in a method of controlling the growing of algae, the method including the steps of controlling with a first control system the spectrum and intensity of a plurality of LED lights for growing algae; monitoring with one or more sensors traits of the algae; changing the spectrum and intensity of the plurality of LED lights while continuing to monitor with the one or more sensors traits of the algae; correlating traits of the algae with the spectrum and intensity of the plurality of LED lights; transmitting with a remote device the spectrum and intensity settings of the plurality of LED lights for a desired trait of the algae to a second control system to grow algae with the desired traits using the second control system.

[0044] Preferably the method includes the step of transmitting from the first control system to the remote device the spectrum and intensity settings of the plurality of LED lights and the correlating sensor data for the traits of the algae. Preferably the step of correlating traits of the algae with the spectrum and intensity of the plurality of LED lights is performed on the remote device.

[0045] Preferably the method further includes correlating traits of the algae with the flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added.

[0046] Preferably the method includes changing one or more of the spectrum and or intensity settings of the plurality of LED lights, flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added to optimise the traits of the algae. Preferably AI is used to change one or more of the spectrum and or intensity settings of the plurality of LED lights, flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added to optimise the traits of the algae. Preferably the method further includes transmitting with the remote device settings related to one or more of flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added to the second control system to grow algae with the desired traits using the second control system.

[0047] In another aspect, the present invention broadly resides in a method of controlling the growing of algae, the method including the steps of controlling with a first control system the spectrum and intensity of a plurality of LED lights for growing algae; monitoring with one or more sensors traits of the algae; changing the spectrum and intensity of the plurality of LED lights while continuing to monitor with the one or more sensors traits of the algae; correlating traits of the algae with the spectrum and intensity of the plurality of LED lights; controlling the spectrum and intensity settings of the plurality of LED lights to correspond to a desired trait of the algae.

[0048] It will be appreciated that traits of the algae can include cell count, growth rate, cell health, algae cell density and or the like.

[0049] The features described with respect to one aspect also apply where applicable to all other aspects of the invention. Furthermore, different combinations of described features are herein described and claimed even when not expressly stated. For example the features described in relation to the control system for growing plant matter can apply to the algae cultivation control system and the control system for growing algae and vice versa.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0050] In order that the present invention can be more readily understood reference will now be

made to the accompanying drawings which illustrate a preferred embodiment of the invention and wherein:

[0051] FIG. **1** is a schematic view of a control system according to an embodiment of the present invention;

[0052] FIG. **2** is a schematic view of a portion of a control system according to another embodiment of the present invention; and

[0053] FIG. **3** is a schematic view of a portion of a control system according to a further embodiment of the present invention.

DETAILED DESCRIPTION

[0054] With reference to FIG. **1** there is shown a control system according to an embodiment of the present invention in the form of an algae cultivation control system **10**. The control system **10** includes a controller **12**.

[0055] The controller **12** controls a plurality of LED lights **14,16,18,20,22**. The controller **12** is adapted to control the spectrum of the LED lights **14,16,18,20,22**. The controller **12** is adapted to control the intensity of the LED lights **14,16,18,20,22**. The controller **12** is adapted to control the flickering frequency of the LED lights **14,16,18,20,22**.

[0056] The LED lights **14,16,18,20,22** provide light to plant matter in the form of algae **24** being grown/cultivated, in a growing medium in the form of water, in a container **26**.

[0057] One or more sensors **28** monitor the growth of the algae and communicate the data to the controller **12**.

[0058] The controller **12** communicates with remote devices in the form of a mobile device **30**, a computer **32** and a server **34**. The controller **12** communicates with the server **34** via the internet **36**.

[0059] The controller **12** is able to communicate the settings of the LED lights **14,16,18,20,22** and the data from the one or more sensors **28** to the mobile device **30**, computer **32** and server **34**. The settings of the LED lights **14,16,18,20,22** in the controller **12** can be set by the mobile device **30**, the computer **32** and/or the server **34**.

[0060] The server **34** can correlate data from the one or more sensors **28** with settings of the spectrum, the intensity and/or the flickering frequency of the LED lights **14,16,18,20,22**. The server **34** can determine the spectrum, the intensity and/or the flickering frequency of the LED lights **14,16,18,20,22** that correspond to desirable data from the one or more sensors **28**. The server **34** can set the spectrum, intensity and/or the flickering frequency values in the controller **12** to the determined values. The mobile device **30**, the computer **32** and/or the server **34** can set which data from the one or more sensors **28** is desirable.

[0061] The server **34** can correlate data for different types of algae being grown and/or different traits of the algae **24** being grown. The mobile device **30**, the computer **32** and/or the server **34** can set the different types of algae being grown, or the traits of the algae being grown.

[0062] The controller **12** can control the spectrum, the intensity and/or the flickering frequency of the LED lights **14,16,18,20,22** individually. In this manner, LED lights **14,16,18,20,22** can define different zones, in which different algae is being grown and/or in which different traits of algae are being cultivated.

[0063] With reference to FIG. **2**, there is shown an algae cultivation control system **100**. The algae cultivation control system **100** includes a first control system in the form of a pilot control system **102**, and a second control system in the form of a user control system **104**.

[0064] The pilot control system **102** includes a controller **106** for controlling the spectrum, intensity and flickering rate settings of the plurality of LED lights, flow rate of growing medium, temperature of growing medium, amount of CO₂ added, amount of O₂ added, and or the type and or amount of nutrients added in the cultivation of algae at **108** and **110**. The controller **106** also monitors sensors (not shown) which provide data on the growth of the algae at **108** and **110**. The controller **106** also monitors the waste products created at **112**. At **114** the grown algae is

processed and at **116** the end product is dispatched. The pilot plant is used to test the effects of different settings of the spectrum, intensity and flickering rate settings of the plurality of LED lights, flow rate of growing medium, temperature of growing medium, amount of CO.sub.2 added, amount of O.sub.2 added, and or the type and or amount of nutrients added in the cultivation of algae at **108** and **110**.

[0065] The data from the sensors and the settings of the spectrum, intensity and flickering rate settings of the plurality of LED lights, flow rate of growing medium, temperature of growing medium, amount of CO.sub.2 added, amount of O.sub.2 added, and or the type and or amount of nutrients added is sent via the internet **136** to a remote device in the form of a server **134**.

[0066] The server **134** processes the data and settings in an optimisation module **140**. The optimisation module **140** uses artificial intelligence **142** to optimise the settings to achieve desired traits of the algae. The server **134** maintains a database of optimised settings for different algae strains and different desired traits. On request through a pilot user console **138**, the server **134** can communicate the updated settings to the pilot control system **102**.

[0067] If a user is growing a known strain of algae using the user control system **104**, they can via a user console **144**, download optimised settings from the server **134** to the user control system **104**, to control the cultivation of algae at **146** and **148**. The user control system also monitors the waste production at **150**. At **152** the grown algae is processed and at **154** the end product is dispatched.

[0068] Data and settings from the user control system **104** is uploaded via the internet **136** to the server **134**. The data and settings are fed into the optimisation module **140** so that the artificial intelligence **142** can use the data and settings to further optimise the settings to achieve desired traits of the algae.

[0069] The server **134** includes a notification module **160** which can notify users and pilot users (not shown) via Short Message Service (SMS) **162**, email **164** or AV **166** to new algae recipes or optimizations, or alert users if settings or data are outside of a predetermined range for the pilot control system **102** or user control system **104** respectively.

[0070] With reference to FIG. 3, there is shown a control system **200**. The control system **200** has a controller **202** which controls multiple sections of an algae cultivation system in the form of an inoculation section **204**, a grow section **206** and an inoculation section **208**.

[0071] Each of the sections **204**, **206**, **208** has a localized controller **210,212,214**. The localized controllers **210,212,214** each control LED lights **220,222,224** respectively. The localized controllers **210,212,214** each control solenoids and pumps **230,232,234** respectively to control the flow rate of growing medium, amount of CO.sub.2 added, amount of O.sub.2 added, and or the type and or amount of nutrients added. The localized controllers **210,212,214** also monitor sensors **240,242,244**. The controller **202** can send setting and sensor data to a remote server (not shown) via the internet **236** via Internet of Things (IoT) messaging system. The controller can also receive IoT messages to control the LED lights **220,222,224** and the solenoids and pumps **230,232,234** via the respective localized controllers **210,212,214**.

[0072] An advantage of the preferred embodiment of the control system includes the ability to control the growth of different types of plant matter. Another advantage of the preferred embodiment of the control system includes the ability to control different traits of plant matter being grown. A further advantage of the preferred embodiment of the control system includes that the controller can set the spectrum, intensity and/or the flickering frequency values of the LED lights in response to desired data values from the one or more sensors.

[0073] While the foregoing has been given by way of illustrative example of this invention, all such and other modifications and variations thereto as would be apparent to persons skilled in the art are deemed to fall within the broad scope and ambit of this invention as is herein set forth.

[0074] Throughout the description and claims of this specification the word “comprise” and

variations of that word such as “comprises” and “comprising”, are not intended to exclude other additives, components, integers or steps.

Claims

1. A method of controlling the growing of algae in a liquid growing medium, the method comprising: controlling, by a first control system, a spectrum and an intensity of a plurality of light emitting diode (LED) lights for growing algae; monitoring, by one or more sensors, traits of the algae; changing the spectrum and the intensity of the plurality of LED lights while performing the monitoring; correlating, on a remote device, the traits of the algae with the spectrum and the intensity of the plurality of LED lights; and transmitting, by the remote device, spectrum and intensity settings of the plurality of LED lights for a pre determined trait in the traits of the algae to a second control system, to grow the algae with the pre-determined trait using the second control system, wherein: the first control system comprises a pilot control system for testing effects of different settings in the spectrum and intensity settings on the traits of the algae; the one or more sensors comprises: a spectrophotometer to determine growth or cell density of the algae in the liquid growing medium, and/or a cell count sensor configured to measure a quantity of cells of the algae in a pre-determined volume of the liquid growing medium; the remote device comprises a computer server, the computer server comprising an optimizer for optimizing, by artificial intelligence, the spectrum and intensity settings based on data produced by the one or more sensors, to achieve the pre-determined trait; and the second control system comprises a user control system for growing the algae with the pre-determined trait based on the optimized spectrum and intensity settings for achieving the pre-determined trait.
2. The method of claim 1, wherein the pre-determined trait is selected from the group consisting of: increased cell count, increased growth rate, increased cell health, increased algae cell density, and combinations thereof.
3. The method of claim 1, wherein the artificial intelligence further optimizes LED light spectrum and intensity settings in the second control system, to achieve the pre-determined trait.
4. The method of claim 3, wherein the first control system is further configured to test one or more effects of one or more parameters on the traits of the algae, wherein the one or more parameters is selected from the group consisting of: different flickering rate settings of the plurality of LED lights, a flow rate of the liquid growing medium, a temperature of the liquid growing medium, an amount of carbon dioxide (CO₂) added to the liquid growing medium, an amount of (O₂) added to the liquid growing medium, a type of nutrients added in the liquid growing medium, an amount of nutrients added to the liquid growing medium, and combinations thereof.
5. The method of claim 4, wherein the optimizer is further configured to process data corresponding to the one or more parameters, and wherein the artificial intelligence optimizes the spectrum and intensity settings based on the data corresponding to the one or more parameters to achieve the pre-determined trait.
6. The method of claim 1, wherein the computer server maintains a database of different optimized spectrum and intensity settings for different strains of the algae and different traits in the traits of the algae.
7. The method of claim 1, wherein the computer server comprises a notifier configured to notify users of either the first control system or the second control system of any new settings in the optimized spectrum and intensity settings.
8. The method of claim 1, wherein the computer server comprises a notifier configured to alert users of either the first control system or the second control system when any of the spectrum and intensity settings or any of the data produced by the one or more sensors are outside of a pre-determined range.
9. The method of claim 1, further comprising: downloading, by a user console, the optimized

spectrum and intensity settings from the computer server to the second control system, to control the growing of the algae with the pre-determined trait.

10. The method of claim 1, wherein the one or more sensors further comprises: a temperature sensor for sensing a temperature of the liquid growing medium; a gas sensor for sensing dissolved gas in the liquid growing medium; a sensor for determining growth of the algae; a pH sensor for measuring pH of the liquid growing medium; a salinity sensor for measuring salinity of the liquid growing medium; a nitrate sensor for measuring a concentration of nitrates in the liquid growing medium; and/or a cell health sensor for measuring health of one or more cells of the algae.

11. The method of claim 1, wherein each of the first and the second control systems comprises a controller, wherein the controller of the first control system is configured to control the spectrum and intensity settings of the plurality of LED lights, and wherein the controller of the second control system is configured to control spectrum and intensity settings of a second set of LED lights.

12. The method of claim 11, wherein each of the controllers is configured to control one or more features selected from the group consisting of: a flow rate of the liquid growing medium, a temperature of the liquid growing medium, an amount of carbon dioxide (CO₂) added to the liquid growing medium, an amount of oxygen (O₂) added to the liquid growing medium, a type of nutrients added in the liquid growing medium, an amount of nutrients added to the liquid growing medium, and combinations thereof.

13. The method of claim 12, wherein each of the controllers is configured to monitor the data produced by the one or more sensors.

14. The method of claim 13, wherein each of the controllers is configured to: receive the spectrum and intensity settings of the plurality of LED lights transmitted from the remote device, and control, based on the received spectrum and intensity settings, (i) one or more pumps, one or more valves, one or more solenoids, and/or one or more heaters, and (ii) the one or more features.
